ODE-4

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Variation of Parameters for Linear ODE

① First solve the homogeneous eqn. (y_h)

② Then take the soln $y = y_h \cdot v$ and solve for v to get the particular solution

Solution of DE

Any relation between dependent and independent variable, when substituted in the DE reduces it into an identity is called solution of that DE.

General Soln

A solution of an ODE which contains as many arbitrary constants as the order of that ODE is called general soln. of that ODE.

⊛ $y = cx$ is G.S of $\frac{dy}{dx} = \frac{y}{x}$

Particular Soln.

It is a solution which is obtained from general soln. by giving some particular values to the arbitary constants occuring in the GS of that ODE

Singular Soln.

A soln. which cannot be obtained from general soln. by giving some particular values to the arbitary constants occuring in the general soln.s is called singular soln.

Examples

① Find a soln. of the differential equation $\frac{dy}{dx} + 2xy = 0$ in the form $y = y_1$. Hence solve $\frac{dy}{dx} + 2xy = 4x$ by the substitution $y = vy_1$

→ $dy/dx + 2xy = 0$ gives $dy/y + 2x\,dx = 0$

Integrating, $\log \frac{y}{c_1} = -x^2$

$\Rightarrow y_1 = c_1 e^{-x^2}$

Now we take, $y = v y_1 = v e^{-x^2}$

$$\frac{dy}{dx} = \frac{dv}{dx} \cdot e^{-x^2} - 2x \cdot e^{-x^2} \cdot v$$

Putting in $dy/dx + 2xy = 4x$ we get

$$\frac{dv}{dx} \cdot e^{-x^2} - 2vx e^{-x^2} + 2xv e^{-x^2} = 4x$$

$$\therefore dv = e^{x^2} \cdot 4x \cdot dx$$

Integrating, $v = 2 \cdot e^{x^2} + c$

$$y = (2 + c e^{-x^2})$$

② Find the soln. of the DE $\frac{dy}{dx} - y\tan x = 0$ in the form $y = y_1(x)$, say. Hence solve, $dy/dx - y\tan x = \cos x$ by the substitution $y = y_1(x)\,v(x)$

(2) Find ... $y = y_1(x)$, say. Hence solve, $dy/dx - y\tan x = \cos x$ by the substitution $y = y_1(x)\,v(x)$

→ $dy/dx - y\tan x = 0 \Rightarrow \frac{dy}{y} = \tan x\,dx$

Integrating $\Rightarrow \ln y = -\ln(\cos x)$

$\Rightarrow y_1 = \sec x$

Now $y = y_1 \cdot v = v\sec x$

$\frac{dy}{dx} \Rightarrow v\sec x\tan x + \frac{dv}{dx}\sec x - v\tan x\sec x = \cos x$

$\Rightarrow \frac{dv}{dx} = \cos^2 x$

$\Rightarrow dv = \cos^2 x\,dx$

Integrating $\Rightarrow v = \int \frac{1}{2}(1+\cos 2x)\,dx + c$

$\Rightarrow v = \frac{x}{2} + \frac{\sin 2x}{4} + c$

$\Rightarrow y = \frac{1}{2}\sec x\left\{x + \frac{\sin 2x}{2} + c\right\}$

(3) Solve by method of variation of parameters

$dy/dx - 5y = \sin x$

Homogeneous eqn: $\frac{dy}{dx} - 5y = 0 \Rightarrow \frac{dy}{y} = 5dx$

$\Rightarrow \ln y = 5x$

$\Rightarrow y_1 = e^{5x}$

To solve the NHDE,

$y = v\,y_1 = v\,e^{5x}$

$\frac{dy}{dx} = e^{5x}\left(5v + \frac{dv}{dx}\right)$

$\Rightarrow e^{5x}\left(5v + \frac{dv}{dx}\right) - 5ve^{5x} = \sin x$

$\Rightarrow dv/dx = \sin x\, e^{-5x}$

$\Rightarrow dv = \sin x\, e^{-5x}\,dx$

Integrating,

$v = I + c$

$I = \int e^{-5x}\sin x\,dx$

$= \left(-\frac{e^{-5x}}{5}\right)\sin x - \int \frac{e^{-5x}}{-5}\cos(x)\,dx$ $\quad\left[\int uv = u\int v + \int v\cdot\frac{du}{dx}\right.$

$= -\frac{\sin x\cdot e^{-5x}}{5} + \frac{1}{5}\int e^{-5x}\cos(x)\,dx$ $\quad\left.\right]$

$= -\frac{\sin x\, e^{-5x}}{5} + \frac{1}{5}\left[-\frac{e^{-5x}}{5}\cdot\cos x - \int \sin x\cdot\frac{e^{-5x}}{5}\,dx\right]$

$\therefore I = -\frac{\sin x\, e^{-5x}}{5} - \frac{1}{25}e^{-5x}\cos x - \frac{1}{25}I$

$\frac{26}{25}I = -\frac{e^{-5x}}{5}\sin x - \frac{e^{-5x}\cos x}{25}$

$\Rightarrow I \Rightarrow \frac{25}{26}\left[-\frac{e^{-5x}}{5}\sin x - \frac{e^{-5x}}{25}\cos x\right] + c_1$

$y = ce^{5x} - \frac{5}{26}\sin x - \frac{1}{26}\cos x$

$\Rightarrow I \Rightarrow \frac{25}{26}\left[-\frac{e}{5}\sin x - \frac{25}{\cdots}\right] + \cdots$

$y = ce^{5x} - \frac{5}{26}\sin x - \frac{1}{26}\cos x$

(4) $\frac{dy}{dt} - 2y = e^{3t}$ [NHDE]

Homogeneous eqn: $\frac{dy}{dt} = 2y \Rightarrow \frac{dy}{y} = 2dt$

$\Rightarrow \ln y = 2t$

$\Rightarrow y_1 = ce^{2t}$

$y = v y_1 = e^{2t} \cdot v$ (1) be the soln of NHDE,

$\therefore$ it will satisfy the previous given DE.

$\therefore \frac{dy}{dt} \Rightarrow e^{2t}\frac{dv}{dt} + 2e^{2t} v - 2e^{2t} v = e^{3t}$

$\Rightarrow \frac{dv}{dt} = e^t \Rightarrow v = e^t$

$\Rightarrow y = e^{3t}$

$y = ce^{2t} + e^{3t}$

(5) $y' - \left(\frac{2}{t+1}\right)y = (t+1)^3$

$\Rightarrow \frac{dy}{dt} - \left(\frac{2}{t+1}\right)y = (t+1)^3 \rightarrow$ NHDE

Homogeneous DE: $\frac{dy}{dt} - \left(\frac{2}{t+1}\right)y = 0$

$\Rightarrow \frac{dy}{y} = \frac{2dt}{t+1}$

$\Rightarrow \ln y = 2\ln(t+1) + c_1$

$\Rightarrow y_h = c(t+1)^2$

Now for the NHDE,

$y = v(t)\cdot(t+1)^2$

$\frac{dy}{dt} \Rightarrow \frac{dv}{dt}(t+1)^2 + 2(t+1)\cdot v - 2(t+1)\cdot v = (t+1)^3$

$\Rightarrow \frac{dv}{dt} = (t+1)$

$\Rightarrow dv = (t+1)dt$

$\Rightarrow v = \frac{t^2}{2} + t$

$\Rightarrow y_p = (t+1)^2\left(\frac{t^2}{2} + t\right)$

Soln: $y = y_p + y_h$

$= c(t+1)^2 + (t+1)^2\left(\frac{t^2}{2} + t\right)$